Vuk Topalovic

Senior Full-Stack Engineer

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PROFILE

Full-stack engineer who builds web platforms end to end, from the data model and API to the interface. Since 2022 I have built EU-funded platforms for energy, buildings and smart cities at European Dynamics, owning features from requirements to deployment, and I publish developer tools that measure their own results.

SKILLS

Frontend	TypeScript, React, Next.js, Angular, React Query, Jest, Testing Library
Backend	Java, Spring Boot, Spring Security, Hibernate, Node.js, Express, MQTT, Keycloak, OAuth2, JWT
Data & infrastructure	PostgreSQL, MongoDB, MySQL, Prisma, Docker, Jenkins, GitHub Actions, Linux
AI tooling	Anthropic SDK, OpenAI SDK, Vercel AI SDK, pgvector, Zod

EXPERIENCE

Full-Stack Software Engineer · European Dynamics — Dec 2022 — Present · Remote (Athens, Greece)

- **OpenBEP4EU.** Full-stack development for an EU LIFE project that turns the ISO 52000 standards into open calculation methods for Energy Performance Certificates and the Smart Readiness Indicator.
- **EnerShare.** Designed and shipped secure REST APIs for a European energy data space (IDSA, GAIA-X) supporting 7 pilot deployments and 11 cross-sector use cases.
- **Resonance.** Built real-time MQTT pipelines that bring sensor data from several cities into one backend, plus marketplace features and services.
- **UnderSec and Tenacity RMT.** Keycloak, JWT and RBAC authentication across Dockerised microservices with Jenkins and GitHub Actions CI/CD; a React and TypeScript risk-management dashboard.

Java · Spring Boot · Node.js · TypeScript · React · PostgreSQL · Keycloak · MQTT · Docker

Software Engineer · Freelance — 2026 — Present · Remote

- **Port appointment booking.** TypeScript and Node.js service that books truck appointments at Saudi land ports (FASAH) for a logistics client's team of drivers.
- Ranks slots by each driver's preferred hours and assigns drivers to distinct slots with min-cost matching (Hungarian algorithm); adaptive concurrency under rate limits. Now porting the core to Rust.

TypeScript · Node.js · Rust

Operations Coordinator · Robert Bosch — Mar 2019 — Dec 2022 · Belgrade, Serbia

SELECTED PROJECTS

token-warden — github.com/vukkt/token-warden

Claude Code plugin that benchmarks agent-memory rules on a frozen test suite and keeps only those that save at least twice what they cost. About 1,180 tests, 96% line coverage. TypeScript · Node.js · SQLite · Zod · Claude Code

iglass — github.com/vukkt/iglass

Liquid Glass theme for iTerm2 and Ghostty on macOS and Linux, using Apple's real glass material in Ghostty. Every color is contrast-checked; 158 tests. Ships via Homebrew. Shell · Python · jq · Homebrew

PR Scout — npmjs.com/package/pr-scout

CLI that scores good-first-issue tickets with an LLM for clarity, scope and spam signals, and filters out stale repositories. Published on npm. TypeScript · Node.js · GitHub API · Groq

Trinomen — npmjs.com/package/trinomen

Three-agent LLM pipeline (router, worker, cross-provider reviewer). Benchmarks showed full review cost about five times the tokens for worse results, so the loop now typechecks first. Node.js · Vercel AI SDK · Zod · SQLite

EDUCATION

BSc, Road Traffic, Faculty of Transport and Traffic Engineering, University of Belgrade — 2014 — 2019

Java Spring Boot Master Class, Amigoscode — 2022

LANGUAGES

Serbian (native)

English (professional working proficiency)